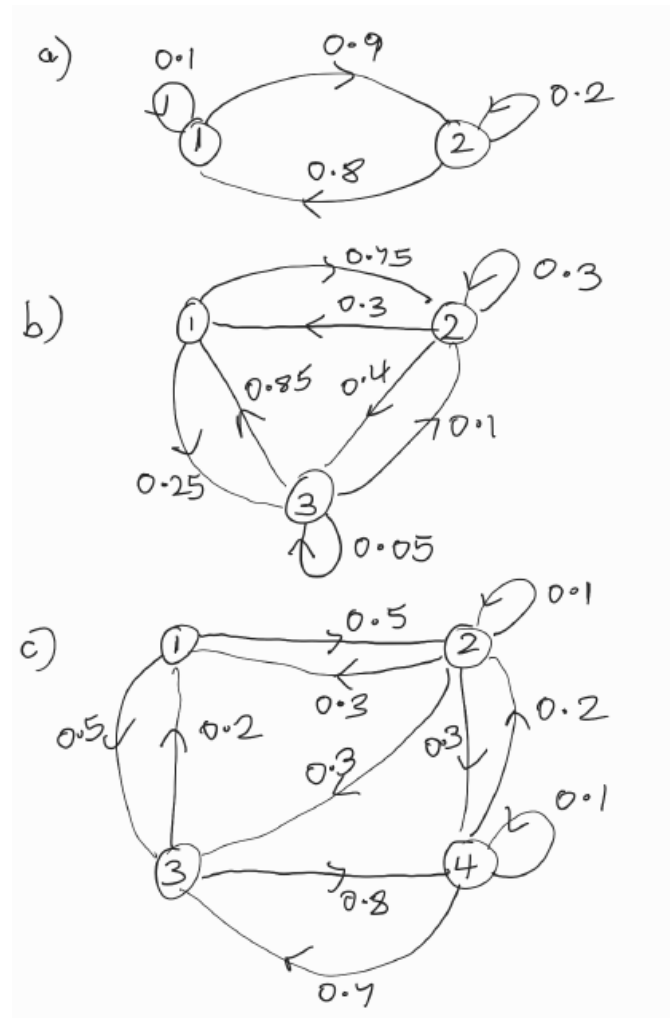


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Q1. State the transition matrices for the following transition diagrams.



Q2. Draw the transitions diagrams for the following transition matrices.

a) $\begin{pmatrix} 0.3 & 0.7 \\ 0.6 & 0.4 \end{pmatrix}$; b) $\begin{pmatrix} 0.1 & 0.9 & 0 \\ 0.25 & 0.25 & 0.5 \\ 0 & 0.45 & 0.55 \end{pmatrix}$; c) $\begin{pmatrix} 0.25 & 0.25 & 0.25 & 0.25 \\ 0.1 & 0.2 & 0.4 & 0.3 \\ 0 & 0 & 0.9 & 0.1 \\ 0 & 0.5 & 0.3 & 0.2 \end{pmatrix}$

Q3. In a certain city, if today is sunny, tomorrow will be sunny 80% of the time. If today is cloudy, tomorrow will be cloudy 60% of the time.

- Construct a transition diagram, clearly labelling the probabilities, to represent this situation.
- Construct a transition matrix to represent this situation.
- If today is sunny, find the probability that it will be cloudy the day after tomorrow using:
 - a probability tree;
 - a Markov chain

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Q4. Shaggy, Scooby Doo and Fred like playing frisbee together. Shaggy throws the frisbee to Scooby Doo $\frac{1}{2}$ of the time and to Fred $\frac{1}{2}$ of the time. Scooby Doo always throws to Fred. Fred throws to Shaggy $\frac{3}{4}$ of the time and to Scooby Doo $\frac{1}{4}$ of the time.

- Construct a transition diagram, clearly labelling the probabilities, to represent this situation.
- Construct a transition matrix to represent this situation.
- If Scooby Doo throws the frisbee initially, what is the probability that Scooby Doo will have the Frisbee after the **second** throw? Use a probability tree and a Markov chain to obtain the answer.
- In the long run, what proportion of the time do each of the players have the frisbee?

Q5. In a small town there are two places to eat, a Chinese restaurant and a pizza restaurant. Everyone in town eats dinner in one of these two restaurants or has dinner at home.

- For those who eat in the Chinese restaurant, 60% eat at home next time, and 20% go to the pizza restaurant.
- For those who eat at the pizza restaurant, 20% will eat there again next time and 70% will eat at home.
- For those who eat at home, 10% go to the Chinese restaurant next time and 10% go to the pizza restaurant.

We are interested in success of these places in terms of their business.

- Construct a transition diagram, clearly labelling the probabilities, to represent this situation. (Use the order 1, 2, 3 = Chinese, Pizza, Home)
- Construct a transition matrix to represent this situation.
- What is the probability that a person who ate in the Chinese will go back to the Chinese 2 weeks later? Use a probability tree and a Markov chain to obtain the answer.
- Determine the long-term probability for each scenario.